

TAKEOFF — Business Plan

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1. What it is

TAKEOFF is a done-for-you plan-measurement service for trades.

A tradie sends us their plan set. We send back a room-by-room m² breakdown the same day. Every number shows its working. Every gap in the drawings comes back as a question, not a guess.

That's the whole product. There is nothing to install, nothing to learn, no seat licence, no onboarding call.

It is a service they send plans to, not software they have to learn. That distinction is the entire positioning, and it is the reason this can win against better-funded products. A solo tiler does not want a measurement tool. He wants the measuring to have already happened.

What comes back:

- Floor m² per room, net of what he doesn't tile
- Wall m² per wall, at true tiling heights — not assumed ceiling heights
- Feature-tile zones priced separately, because they're a different SKU
- Niches, skirting, linear metres
- Wastage toggle: measured / +10% / +15%, side by side, his choice
- A **QUESTIONS FOR YOU** section leading the document
- A **CONFIRM BEFORE QUOTING** checklist closing it

Delivery: PDF or markdown, back by end of day. Intake by email or WhatsApp — wherever he already is.

2. The problem

Solo tradies lose 45+ minutes per quote measuring plans at night.

The quoting happens after the tools go down. Six on site, then dinner, then a set of PDFs on a laptop at the kitchen table with a scale rule and a calculator, working out how many square metres of wall tile are in a bathroom. It's the least enjoyable hour of the day and it's unpaid.

And it's high-stakes:

- **One wrong wall eats the job's margin.** Measure a 2700 ceiling where the drawing calls a 450 splashback and you've quoted 29 m² of tiling that doesn't exist — or the reverse, and you wear the difference.
- **A bounding-box read of a non-rectangular room is catastrophic.** In our sample set, the master ensuite measures 19.9 m² as a rectangle and 10.7 m² as a tiled floor. That's a 9 m² over-order — roughly \$1,400 of porcelain plus the labour hours quoted against area that isn't there.
- **The error is invisible until the tiles run out.** By then he's on site, short, and eating it.

Our first customer — a Sydney tiler, and the source of the sample set in this repo — put it plainly:

"It's a pain in the arse for all tradies."

All tradies. Not "for me." He volunteered the market size in the same breath as the complaint. That is the sentence this business is built on.

The competitive nuance underneath: he is not looking for a tool. He has been offered tools. He wants the problem to go away.

3. Why it wins

The market splits into two camps and neither serves a solo tradie well.

Camp 1 — DIY takeoff software (Groundplan, Metres.ai, Bluebeam, PlanSwift). Genuinely good products. They ask the tradie to learn software, set up a project, calibrate a scale, trace areas with a mouse, and maintain a subscription. That is a *new evening task* replacing an old one. Adoption among solo operators is poor for exactly this reason. They sell to the tradie who wants to become an estimator. Most don't.

Camp 2 — human estimating services. Accurate, trusted, and priced for builders and commercial contractors — typically several hundred dollars and multi-day turnaround. Structurally unable to serve a \$6k bathroom quote where the tradie needs an answer tonight.

TAKEOFF sits in the gap: the outcome of camp 2 at a price and speed that works for camp 1's customer.

Three things make that possible:

3.1 The AI pipeline runs on an existing subscription

The marginal cost of a takeoff is close to zero. Deterministic extraction (PyMuPDF) plus a model doing the one thing models are actually good at here — associating printed dimensions with the elements they describe — running on a subscription that's already paid for. That is what makes **per-job pricing** viable at a price a solo tradie will pay without thinking about it. A human estimator cannot get there. A software company charging per seat has no reason to try.

3.2 Distribution nobody else has

The founder talks to tradies face-to-face six days a week.

This is not a growth hack, it's the actual moat on the demand side. The hardest thing about selling to trades is that they don't answer cold email, don't read LinkedIn, don't attend webinars, and don't trust software companies. They trust the person standing in front of them who knows what a nib wall is.

An incumbent with 50× the funding still has to buy Google ads against "tile takeoff software" and convert a stranger. We get a warm conversation on a job site, an existing relationship, and the ability to hand back a finished takeoff for a job the tradie is quoting *this week*. Customer #1 came from that. So will customers #2 through #5.

3.3 We compete on accuracy, and we can prove it

Everyone claims accuracy. We are the only ones who will hand over the working, list what we weren't sure about, and refuse jobs we can't measure properly. Section 4 is the mechanism.

4. The accuracy doctrine (the moat)

Accuracy is the product. Not speed, not price, not the interface. If the numbers are wrong, nothing else matters, and if they're reliably right, nothing else has to be perfect.

Six commitments, each of them a real mechanism already built into this repo:

4.1 Intake requirements, enforced

Bad input is the largest single source of bad output, and it is the cheapest to eliminate. Every job passes an

automated gate (`INTAKE.md`, `takeoff.py`) **before** any analysis: vector PDF with a real text layer, printed mm dimensions, elevations if walls are wanted, notes pages included, plus three answers from the tradie.

Fail → a polite `REJECTED_<job>.md` saying exactly what's missing and what to send instead. No partial takeoff, no best-effort guess.

Turning work away looks like lost revenue. It is the opposite: it is the only way to make "our numbers are right" a claim we can keep. And "I can't measure this yet, here's what I need" is a message no competitor sends, which is precisely why it builds trust.

4.2 Deterministic extraction and deterministic math

The model never does arithmetic that matters and never reads a number off a picture.

- Text and coordinates come out of the PDF with PyMuPDF. Deterministic.
- Areas are computed by explicit formulas from those extracted values. Deterministic.
- **We measure from stated dimensions only. We never scale off the drawing** — a plan printed "fit to page" is no longer at its stated scale, and scaling it produces numbers that look right and are 3–8% wrong.

4.3 AI is used for exactly one thing

Associating a printed dimension with the element it describes: *which wall is this 2150 the height of? is this a wall dimension or a fixture centreline? does this "450" mean a splashback or a cabinet?*

That is a judgement task on a drawing, which is where a model genuinely outperforms a rule. It is not asked to do arithmetic, not asked to invent a missing number, and not asked to decide anything it can't point at on a sheet.

4.4 A cross-check engine, not a spot check

Six checks on every job (`TAKEOFF_METHOD.md` §5), all logged with pass/fail in the delivered document:

1. Segment chains sum to their stated totals
2. Opposite elevations of the same room agree
3. **Wall runs reconcile with the floor perimeter** — the strongest check available, because it ties two independently drawn sheets together
4. Height chains reconcile within each elevation
5. Fixture dimensions agree between plan and elevation
6. Totals fall inside sanity ranges

On the sample set, check 3 came back at 0 mm error on three rooms and 11 mm on a 12-sided cross-shaped ensuite. That's not a claim of accuracy; it's a demonstration the customer can read.

Every room then carries a mechanical **confidence rating** — **HIGH** / **MED** / **LOW** — derived from which checks passed. A LOW room is flagged as not-quotable until its questions are answered.

4.5 Never guess — ask

Every inferred value becomes a **numbered question** at the top of the document, phrased so a tradie can answer it in one word. Not "assumed window height 1360mm" buried in a footnote, which can only be discovered on site. Instead: *"the window height isn't dimensioned on sheet 7.02 — I've used 1360 and deducted 1.36 m². What's the actual height?"*

The first can be answered before the quote goes out. The second can only be discovered after.

Asking beats assuming. That is the brand. In practice it also converts: a supplier who tells you what they don't know is the one you believe about what they do.

4.6 Human sign-off before delivery

Nothing goes to a customer unread. The founder reviews every takeoff before it's sent — scanning the cross-check log, the confidence ratings and the questions list — for as long as volume allows. Sign-off is what makes it a service rather than an API, and it's where we learn what the checks are still missing.

4.7 Accuracy benchmarked by customer red-pen rounds

The only accuracy metric that counts: **give the tradie a red pen and ask him to mark every number he thinks is wrong.**

Every pilot delivery asks for it. Every correction becomes a test case and, where it generalises, a new rule in `TAKEOFF_METHOD.md`. Tracked as:

```
accuracy = 1 - (corrected m² ÷ total m² delivered)
```

Target: **≥98% after one correction round by pilot #5**. Published to customers. If it can't hold that, section 8 applies.

5. Offer and pricing

Per-job pricing, set by discovery. We ask what it's worth before we tell them what it costs — customer #1's brief is literally *"mark the errors, and name your price."*

Placeholder tiers, to be replaced with real numbers after five discovery conversations:

Tier	Scope	Placeholder	Turnaround
Single room	One wet area, floor + walls	\$__	Same day
Whole reno	3–6 rooms, floor + walls + niches + skirting	\$__	Same day
Full house	Whole set, multi-trade breakdown	\$__	24–48 h

Pricing principles:

- **Per job, not per seat.** He pays when he quotes. No subscription to forget about.
- **Anchor against the alternative**, which is 45 minutes of his evening plus the risk of a wrong wall — not against software seat prices.
- **Never price below the value of the risk removed.** Cheap signals unreliable in trades.
- **Re-issue after answered questions is free.** Charging for the second pass would punish the exact behaviour the accuracy doctrine depends on.

The pilot offer

The first 5 customers are free. Not a discount — free, explicitly and with a name: *the pilot*.

In exchange, two things:

1. **Marked-up corrections.** Red pen on the delivered takeoff, every number he thinks is wrong.
2. **A testimonial**, if the numbers stand up.

Why free is right here: it removes every objection at once, buys the highest-quality training data available (a professional's corrections on real drawings), and makes the ask for corrections *fair* rather than cheeky. Five is enough to find the systematic errors and few enough that it stays a pilot rather than a habit.

6. Go-to-market, play by play

Phase 0 — This week

Send the sample takeoff to **customer #1** (the Sydney tiler whose plans are in this repo). Two questions, nothing else:

1. **Mark every number you think is wrong.**
2. **What would you have paid for this?**

That's the whole message. No pitch, no deck, no pricing page. He already said it's a pain in the arse; this is the thing that makes it stop being one.

Success = a marked-up document comes back. Not a sale. A marked-up document tells us whether the product works. A polite "looks good, mate" tells us nothing — if that's what comes back, push once for specifics.

Phase 1 — 5 free pilots

Source, in priority order:

1. Existing client network — people who already know and trust the founder
2. Face-to-face, on site, six days a week
3. Cold outreach as the top-up, not the strategy

The in-person ask (the one that actually converts):

"How long do you spend measuring plans off for a quote? ... Right. Send me the PDF for the next one and I'll do it for free — you tell me if I got it wrong. No catch, I'm trying to work out if this is worth building."

Cold SMS:

Hey [Name] — [Referrer] gave me your number. I do plan takeoffs for tilers — you send the PDF, I send back the m² per room same day, all the working shown. Doing 5 free while I get it right. Want me to do your next quote? — [Name]

Cold Instagram / Facebook DM (tradies live on Instagram):

Saw your work on the [suburb] bathroom — nice job. Quick one: how long do you spend measuring plans for a quote? I've built something that does the takeoff off the PDF — m² per room, walls, niches, the lot, back same day. Giving away 5 free to get it right. Want yours done?

Follow-up if no reply after 3 days (once, then stop):

No worries if it's not for you — if you've got a plan set sitting there for a quote this week, send it through and I'll knock it over free. Takes me minutes, saves you an evening.

Delivery ritual for every pilot — this is what turns a delivery into learning:

1. Send the takeoff
2. *"Grab a red pen — mark anything you reckon is wrong. Even if you're not sure."*
3. Log every correction as a test case
4. Fix the method, re-issue free
5. Ask: *"what would you have paid for that?"*

Phase 2 — Switch pricing on

Trigger: **accuracy ≥98% after one correction round, across at least 3 pilots.**

- Set real prices from what the five pilots said they'd pay. Take the median, not the top.
- Go back to the pilots first: *"It's live now — \$X a job. Want me to keep doing yours?"* A pilot who converts is worth more than a new lead, and their answer is the real pricing validation.

- **Bake the referral ask into every single delivery.** Not a campaign, a line at the bottom of the document:

"Know another tradie who hates measuring plans at night? Send them my way — their first one's on me."

Trades are a referral market. This is the only distribution that compounds.

Phase 3 — Front door and scale

- **Dedicated intake.** A real email address (plans@...) and a WhatsApp number. Tradies use WhatsApp; meet them there. Auto-acknowledge on receipt with the three intake questions.
- **Scale outreach** on whatever channel actually produced pilots — not all of them.
- **Weekly metrics**, reviewed at the Sunday gate:

Metric	Definition	Target by day 30
Accuracy after corrections	$1 - (\text{corrected m}^2 \div \text{delivered m}^2)$	$\geq 98\%$
Turnaround time	plans received → takeoff delivered	< 8 working hours
Pilot → paid conversion	pilots who pay for a second job	$\geq 40\%$
\$ per job	revenue ÷ jobs delivered	$\geq$ placeholder tier
Intake rejection rate	rejected ÷ submitted	tracked, not targeted
Questions per job	avg. items in QUESTIONS FOR YOU	trending down

7. Unit economics — template

Fill after discovery. Formulas are fixed; the numbers are deliberately blank because inventing them now would be exactly the guessing this business exists to avoid.

Inputs

	Symbol	Value
Price per job	P	\$_____
Jobs per week	J	_____
Minutes per job (review + send)	M	_____
Weeks per month	W	4.33
Model / infra cost per job	C_v	\$_____
Fixed monthly cost (subscription, tools)	C_f	\$_____
Founder's target hourly rate	R	\$_____/h

Formulas

Monthly revenue	$= P \times J \times W$
Monthly variable cost	$= C_v \times J \times W$
Monthly gross profit	$= (P - C_v) \times J \times W$
Gross margin %	$= (P - C_v) \div P$
Monthly hours	$= (M \div 60) \times J \times W$

Effective hourly rate = Monthly gross profit ÷ Monthly hours
 Monthly net = Monthly gross profit - C_f
 Break-even jobs/month = C_f ÷ (P - C_v)
 Capacity ceiling (jobs) = available hours/month ÷ (M ÷ 60)

The three numbers that decide it

1. **Effective hourly rate** — if this is below what the founder earns on the tools, the business is a hobby regardless of revenue.
2. **Break-even jobs/month** — if this is more than a handful, pricing is wrong.
3. **Capacity ceiling** — the point where human sign-off (\$4.6) has to change. Know it before hitting it, and decide then whether to hire a reviewer or narrow the checks.

Sensitivity to fill in after pilots

P	J/wk	Monthly revenue	Monthly hours	Effective \$/h
\$__	__			
\$__	__			
\$__	__			

8. Risks and kill criteria

Written down in advance, because the point of a kill criterion is that it's decided while you're still objective.

Risk 1 — Accuracy fails after one correction round

The signal: a pilot marks up a takeoff, we fix it and re-issue, and he *still* finds material errors. Or errors are idiosyncratic rather than systematic — meaning there's no rule to learn, just noise.

Why it kills it: the entire proposition is that these numbers are trustworthy. A takeoff that needs checking is worse than useless — the tradie now does his own measuring *and* reads ours.

KILL: if accuracy is below 95% after one correction round on 3 of the first 5 pilots.

Mitigation before then: narrow the scope. Floors only, rectangular rooms only, one trade. A small guarantee kept beats a broad one broken.

Risk 2 — Tradies won't pay

The signal: pilots love it, use it, praise it — and go quiet when pricing turns on. Or they name a price that doesn't clear the effective hourly rate in \$7.

Why it might happen: the pain is real but intermittent. A tradie quoting two jobs a month may absorb 90 minutes rather than open his wallet. Free changes behaviour in ways that don't survive a price.

KILL: if fewer than 2 of the first 5 pilots pay for a second job within 30 days of pricing going live.

Mitigation before then: test pricing earlier — ask "what would you pay?" at pilot #2, not pilot #5. Test a per-job price against a small monthly retainer for high-volume quoters.

Risk 3 — Real-world plans are too poor for intake

The signal: a high proportion of submitted sets fail the gate. Photos of plans, raster scans, undimensioned sketches, floor plans with no elevations. The addressable market is then much smaller than the number of tradies who have the problem.

This is the most likely failure mode, and it's the one the sample set already hints at: a professionally-produced architectural pack from a Sydney design studio still arrived missing a floor plan sheet, with an undimensioned ceiling height, three undimensioned windows, no door heights, and two chains that contradict each other by 165 mm. If *that's* the good end of the distribution, the middle is rough.

KILL: if more than 60% of genuinely-submitted sets fail intake, and tradies won't or can't get better files from their designers.

Mitigation before then: go upstream. Sell to the designers and builders who *produce* the sets, or partner with them so the plans arrive measurable. That's a different business — but it's a real one, and this failure mode points straight at it.

Risk 4 — Platform dependency (watch, not kill)

The pipeline runs on a third-party model subscription. Pricing or terms could change. **Watch:** cost per job monthly. **Mitigate:** the extraction layer is already deterministic and portable; only the association step is model-dependent, and it is model-agnostic in principle.

The Sunday gate

Every Sunday, one hour. Four questions, in order:

1. What did the metrics do this week? (§6 Phase 3 table)
2. Did any kill criterion trigger?
3. What did customers actually say — quoted, not paraphrased?
4. What's the single most important thing to do next week?

If a kill criterion has triggered, it dies without ceremony. No pivot in the same meeting, no "let's give it two more weeks." The criterion was set when we were objective; honour it. Write down what was learned and stop.

This is the discipline the whole plan rests on. *If it can't hold numbers, it doesn't deserve to exist* — that's true of the product and it's true of the business.

9. The first 30 days

Built around weekday evening blocks (roughly 90 minutes, after the tools) and a Sunday review. Nothing here requires quitting anything.

Week 1 — Prove it on real plans

Day	Block
Mon	Finalise the sample takeoff. Read it as a tiler would, not as a builder.
Tue	Send to customer #1. Two questions: mark the errors, name your price.
Wed	Draft the intake email/WhatsApp template and the three questions.
Thu	List 15 tradies in the existing network. Names, trade, how you know them.
Fri	Chase customer #1 gently if quiet. Prep the in-person ask (§6).
Sat	On-site conversations. Ask the 45-minute question. Say nothing about software.
☉ Sun	Gate 1. Did a marked-up takeoff come back? What was wrong, and was it systematic?

Week 2 — Pilots 1–3

Day	Block
Mon	Fix every correction from customer #1. Re-issue free. Add each as a test case.
Tue	Send 8 cold SMS/DMs from the list of 15.
Wed	Run pilot #2's plan set end to end. Time it — that's the real turnaround number.
Thu	Deliver pilot #2 with the red-pen ask. Follow up the 8 messages once.
Fri	Run and deliver pilot #3.
Sat	Face-to-face. Two more pilots booked. Ask every one what they'd pay.
☉ Sun	Gate 2. Accuracy so far? Turnaround? Any kill criterion triggered?

Week 3 — Pilots 4–5, and pricing discovery

Day	Block
Mon	Fold week-2 corrections into <code>TAKEOFF_METHOD.md</code> . This is the compounding asset.
Tue	Run pilot #4.
Wed	Deliver #4. Explicit pricing conversation: <i>"what's this worth to you?"</i>
Thu	Run and deliver pilot #5.
Fri	Collect all five pricing answers. Take the median. Set the three tiers.
Sat	Ask the two happiest pilots for a testimonial and one referral each.
☉ Sun	Gate 3 — the big one. Accuracy $\geq 98\%$? If yes, pricing goes live Monday. If no, fix or kill.

Week 4 — Switch pricing on

Day	Block
Mon	Pricing live. Message all five pilots: <i>"it's \$X a job now — want me to keep doing yours?"</i>
Tue	Set up the front door: <code>plans@</code> address, WhatsApp number, auto-acknowledge with the three questions.
Wed	Add the referral line to the delivery template. Work the referrals from week 3.
Thu	First paid job, start to finish. Time it. Record the true cost per job.
Fri	Fill in §7 with real numbers. Compute the effective hourly rate.
Sat	Face-to-face outreach on the channel that actually produced pilots.
☉ Sun	Gate 4 — the 30-day verdict. Conversion $\geq 40\%$? Effective \$/h clears the bar? Continue, narrow, or stop.

The 30-day verdict

Three numbers decide it. Everything else is commentary.

	Continue if	Kill if
Accuracy after one correction round	$\geq 98\%$	$< 95\%$ on 3 of 5

Pilot → paid within 30 days of pricing	≥ 2 of 5	< 2 of 5
Effective hourly rate	> on-the-tools rate	< on-the-tools rate with no path up

TAKEOFF measures from stated dimensions only. It never scales off the drawing, and it never guesses. That rule is the product, and it is the business.